DevWatchAI

Senior Software Engineer (SDE) Interview & Architecture Deep-Dive Guide

SDE-II / Senior SDE Prep Target Workspace: `DevWatchAI` Role: Senior Software Engineer & Interviewer

DevWatchAI - Senior Software Development Engineer (SDE) Interview & Architecture Deep-Dive

1. PROJECT OVERVIEW

What Problem This Project Solves

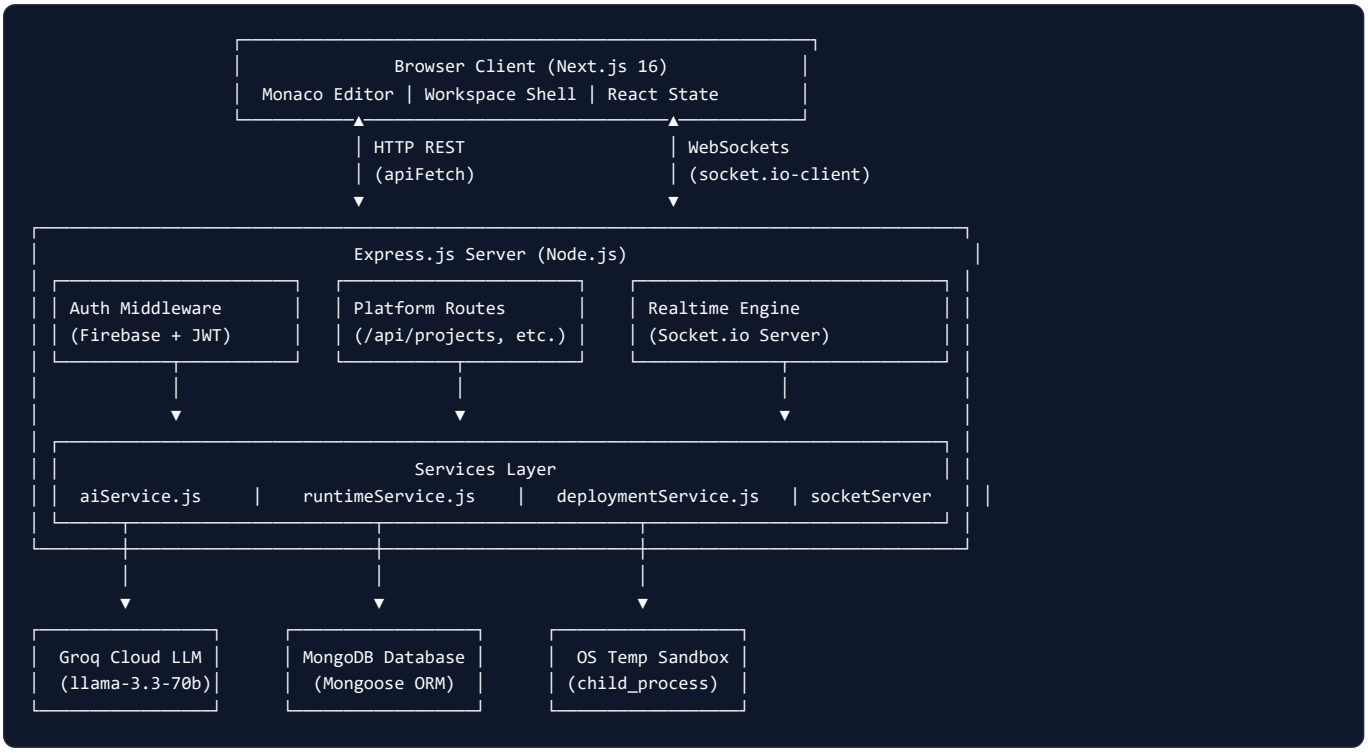
DevWatchAI is an AI-native, collaborative cloud IDE and real-time developer workspace. It bridges the gap between local code execution, collaborative pair programming, and continuous AI quality assurance. Standard developer workflows suffer from fragmented tooling: code is written in a local IDE, manually submitted for code review, tested in separate sandboxes, and monitored across disconnected dashboards.

DevWatchAI unifies these workflows into a single browser-based environment:

- **Instant Cloud Workspace**: Provision multi-file project environments with Monaco-based editing without local environment setup.
- **Real-Time Collaboration**: Google Docs-style live co-editing, active file presence, cursor tracking, and inline workspace chat.
- **AI-Powered Engineering Co-Pilot**: Deep code reviews, automated bug detection, automated fix generation, and context-aware chat assistant powered by high-speed Groq LLMs (`llama-3.3-70b-versatile`).
- **In-Browser Code Execution Engine**: Sandboxed node/python/tsx code execution with streaming logs and line-level error tracking over WebSockets.
- **Build & Artifact Generation**: One-click project packaging into exportable ZIP archives and deployment tracking.
- **Workspace Telemetry**: Real-time CPU/RAM usage logging, latency recording, error tracking, and audit logging.

Architectural Blueprint & Design Rationale

DevWatchAI adopts a **decoupled Client-Server architecture** utilizing Next.js for the client-side single-page application and a dedicated Node.js/Express server for API endpoints, WebSocket persistence, process management, and MongoDB interactions.



Why This Architecture Was Chosen:

1. **Separation of Concerns**: Decoupling the frontend (Next.js rendering UI) from the backend execution engine allows independent scaling of memory-heavy process runners from light static web serving.
2. **WebSocket & Long-Polling Support**: Node.js event-driven non-blocking I/O handles thousands of persistent WebSocket connections efficiently for live presences and real-time streaming execution logs.
3. **Low-Latency AI Processing**: Interfacing directly with Groq's high-speed LPU infrastructure via Server-to-Server API calls hides API keys from the browser client and enforces JSON schema responses.
4. **Sandboxed Subprocess Spawning**: Node's native `child_process.spawn` allows spawning isolated Node, TSX, and Python runtimes in temporary OS directories, streaming stdout/stderr back in real time.

Complete System Workflow (User Action → Response)

Workflow Example: User Clicks "Run Project" in the IDE

1. **User Action**: User clicks the "Run Code" button inside cloud-ide.tsx.
2. **Frontend Request**: The `runProject` handler calls `apiData('/api/projects/:projectId/run', { method: 'POST' })`.
3. **HTTP Processing**:
   - Request reaches Express server (`server.js`).
   - `requireAuth` middleware validates the Bearer token in the `Authorization` header, looks up the `User` in MongoDB, and attaches `request.user` .
   - Request hits router handler in extendedRoutes.js.
   - `assertProjectAccess` checks if the user belongs to the workspace associated with the project.
4. **Backend Processing**:
   - `startExecution` in runtimeService.js creates an `Execution` record in MongoDB (`status: 'queued'`).
   - Creates an isolated temp directory using Node's `fs.mkdtemp(path.join(os.tmpdir(), 'devwatch-run-'))` .
   - Iterates through project files in MongoDB and writes them to the temp directory via `fs.writeFile` .
   - `selectRunner` inspects the project files to find an entry point (`package.json` main, `index.ts` , `main.py`) and returns binary details (`node` , `tsx` , or `python`).
   - Spawns process using `child_process.spawn` with stdout/stderr listeners.
5. **Real-time Log Streaming**:

- As stdout/stderr output arrives, `runtimeService.js` emits Socket.io events (`execution:log`) to the room `project:projectId` .
- The frontend's `socket.io-client` listener receives log chunks and appends them into the console terminal state in real time.

6. **Finalization & Response**:
   - HTTP POST `/run` responds immediately with `202 Accepted` containing initial execution metadata.
   - Upon process exit, `finalizeExecution` updates MongoDB `Execution` status to `success` or `failed` , logs duration, creates `Activity` and `Monitoring` records, cleans up temp directory with `fs.rm` , and emits `execution:updated` over Socket.io.

Main Features

1. **Multi-Tenant Workspace & Project Hierarchy**: Workspaces contain projects; projects contain files, versions, deployments, monitoring records, and chats.
2. **Monaco Editor Integration**: Full syntax highlighting, diff viewing, line number mapping, and auto-formatting across 18+ file extensions (`.ts` , `.tsx` , `.js` , `.py` , `.json` , etc.).
3. **AI Engineering Suite**:
   - **Code Review**: Analyzes code quality, maintainability, security, and performance scores (0-100) with line-specific issue callouts.
   - **Bug Detection**: Detects edge cases, potential crashes, and suggests reproduction steps and unit test strategies.
   - **Auto-Fix**: Generates drop-in code fixes with risk analysis.
   - **Workspace Chat Assistant**: Q&A over project code context.
   - **Full Project Analysis**: Comprehensive architectural evaluation across all repository files.
4. **Real-time Code Execution Sandbox**: Spawns isolated process runners, streams stdout/stderr, parses stack traces for file/line errors, and auto-terminates after a 30-second timeout.
5. **Real-time Presence & Active Cursors**: Socket.io room membership tracks active viewers, current open file, active cursor position, and typing indicators.
6. **Version Control & File History**: Automatic snapshot creation on file create, rename, move, update, import, and delete, with one-click revision rollback.
7. **Workspace Team Invites**: Resend-powered transactional email invites with cryptographically hashed tokens (`sha256`), role assignment, and single-click acceptance.
8. **Exportable Project Packaging**: In-memory ZIP stream generation using `archiver` for instant project downloads.

Real-World Production Use Cases

- **Cloud Developer Platforms**: Providing instant dev environments without local machine setup (e.g., StackBlitz, Replit).
- **AI Code Review Automation**: Replacing or supplementing manual GitHub Pull Request reviews with instant security & code smell audits.
- **Coding Interview & Assessment Platforms**: Conducting technical interviews with real-time pair programming, live code execution, and AI-assisted candidate evaluation.
- **Internal Micro-App Prototyping**: Enabling engineers and non-technical stakeholders to build, run, and share Node.js and Python prototypes rapidly.

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2. TECH STACK DEEP DIVE & INTERNAL MECHANICS

Complete Technology Matrix

Category	Technology	Version	Purpose in DevWatchAI	Alternative
Frontend Framework	Next.js (App Router)	`16.2.6`	Client-side application routing, page rendering, static asset optimization	Vite + React SPA
UI Library	React	`18.3.1`	Component-based interactive UI state management	Vue.js, Svelte
Code Editor	`@monaco-editor/react`	`^4.7.0`	In-browser code editing engine (VS Code editor core)	CodeMirror, Ace
Realtime Transport	Socket.io / Socket.io-client	`^4.8.1`	Bi-directional WebSocket communication for live presence and logs	Server-Sent Events (SSE), WebRTC
Styling Engine	Tailwind CSS / PostCSS	`^3.4.17`	Utility-first CSS styling system	Styled Components, Emotion
Icons & Motion	Lucide React / Framer Motion	`0.468.0` / `11.15.0`	UI iconography and smooth modal/drawer animations	FontAwesome, React Spring
Charts	Recharts	`^2.15.0`	Visualizing CPU, RAM, latency, and AI usage metrics	Chart.js, D3.js
Backend Engine	Express.js	`^4.21.2`	RESTful API routing, middleware execution, HTTP server	Fastify, NestJS
Database & ORM	MongoDB / Mongoose	`^8.9.2`	Document data storage and schema modeling	PostgreSQL + Prisma
LLM Provider	Groq SDK (`groq-sdk`)	`^1.2.0`	Ultra-fast LLM inference (`llama-3.3-70b-versatile`)	OpenAI API, Anthropic Claude
Authentication	Firebase / Firebase-Admin	`^11.1.0` / `^13.0.2`	Client-side OAuth/Password auth & server-side ID token verification	Auth0, NextAuth.js, Clerk
Session Management	JSON Web Token (`jsonwebtoken`)	`^9.0.2`	Signed JWT session token issuance and verification	Express-Session (Redis)
Packaging & Zip	Archiver (`archiver`)	`^7.0.1`	Streaming zip archive creation for project downloads	JSZip
Validation	Zod (`zod`)	`^3.24.1`	Type-safe schema validation (used in TypeScript server mode)	Joi, Yup

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Detailed Technology Deep Dives

1. Next.js 16 & React 18

- **Why used**: Next.js provides modern React bundling, fast hot-reloading, and optimal asset delivery.
- **Internal Mechanics**: The app uses Next.js App Router (`app/layout.tsx` , `app/page.tsx`). All major components use the `"use client"` directive because DevWatchAI is an interactive SPA heavy on browser state (Monaco editor, WebSockets, Canvas/SVG charts).
- **Hooks Used**: `useState` , `useEffect` , `useRef` , `useMemo` , `useCallback` .
- **SSR Handling**: Monaco editor relies on browser `window` APIs (DOM, Canvas, Web Workers). Next.js dynamic import (`dynamic(() => import('@monaco-editor/react'), { ssr: false })`) disables SSR to prevent window hydration mismatch errors.

2. Express.js Backend Framework

- **Why used**: Express provides minimal, high-performance middleware-driven HTTP routing.
- **Middleware Lifecycle Flow**:
  1. `cors` : Validates origin against `process.env.CLIENT_ORIGIN` whitelist.
  2. `express.json({ limit: "10mb" })` : Parses JSON request payloads up to 10MB (necessary for full project imports).

3. `requireAuth`: Custom middleware in auth.js. Extracts `Authorization: Bearer <token>`, verifies JWT signature, fetches user document from MongoDB via `User.findById`, and attaches `request.user`.
4. Route Controller execution inside `asyncHandler` wrapper.
5. Global Error Middleware: Catches uncaught errors and returns `{ message: error.message }` with appropriate status code.

3. Socket.io Real-time Infrastructure

- **Why used**: HTTP polling is inefficient for high-frequency updates like live typing cursors and streaming process output logs.
- **Internal Mechanics**:
  - Attached directly to the Node `http.Server` instance.
  - Connection middleware verifies JWT session token passed via `socket.handshake.auth.token` or headers.
  - Client joins project-specific channels via `socket.join('project:projectId')`.
  - In-memory presence map (`projectPresence = new Map()`) maintains active socket IDs, user metadata, open file ID, cursor coordinates, and typing state.
  - Events emitted: `presence:list`, `file:change`, `file:save-status`, `chat:typing`, `chat:message`, `execution:log`, `execution:updated`, `deployment:updated`.

4. Groq SDK & LLM Engine (`llama-3.3-70b-versatile`)

- **Why used**: Groq's custom LPU (Language Processing Unit) architecture delivers inferencing speeds over 500 tokens/sec, enabling near-instant AI code reviews and chat responses.
- **Structured JSON Mode**: To prevent LLM hallucination and ensure predictable JSON parseability, groqClient.js uses Groq's `response_format: { type: "json_object" }` flag combined with strict system prompts enforcing an `expected_schema`.
- **Fallback JSON Extraction**: `extractJson()` uses regex `/\{[\s\S]*\}/` to extract structured JSON if the LLM wraps response text in markdown block indicators.

5. MongoDB & Mongoose ORM

- **Why used**: Flexible JSON-document schema design fits arbitrary directory file trees, nested AI reviews, execution logs, and dynamic activity metadata without complex SQL migrations.
- **Connection Configuration**: db.js configures `mongoose.connect` with `strictQuery: true`, `connectTimeoutMS: 10000`, `serverSelectionTimeoutMS: 10000`, and `maxPoolSize: 10`.

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3. COMPLETE ARCHITECTURE EXPLANATION & FILE INDEX

Directory & File Mapping

```
DevWatchAI/
├── config/                    # Backend Configuration Modules
│   ├── db.js                  # Mongoose MongoDB connection pooling logic
│   └── firebaseAdmin.js       # Firebase Admin SDK initialization & token verification
├── middleware/                # Express Request Middlewares
│   └── auth.js                # JWT Session signing and Bearer token verification
├── models/                    # Mongoose Schema Definitions (14 Collections)
│   ├── AIReview.js            # AI code analysis & score storage
│   ├── Activity.js            # Workspace audit logging
│   ├── Chat.js                # Project channel chat messages
│   ├── Deployment.js          # Build & deployment tracking records
│   ├── Execution.js           # Subprocess execution history & log storage
│   ├── File.js                # Virtual file system nodes (files and folders)
│   ├── Invitation.js          # Workspace invite tokens & status
│   ├── Member.js              # Workspace membership & RBAC roles
│   ├── Monitoring.js          # System metric time-series snapshots
│   ├── Notification.js        # User alerts & notifications
│   ├── Project.js             # Project container schema
│   ├── User.js                # User profile & onboarding status
│   ├── Version.js             # File revision history snapshots
│   ├── Workspace.js           # Workspace root container schema
│   └── index.js               # Centralized model exports
├── routes/                    # Primary Express REST API Routes (CommonJS)
│   ├── aiRoutes.js            # AI endpoints (/review, /suggestions, /auto-fix, /chat, etc.)
│   ├── extendedRoutes.js      # Advanced endpoints (imports, executions, deployments, invites)
│   └── platformRoutes.js      # Core CRUD endpoints (auth, workspaces, projects, files)
├── services/                  # Business Logic & Infrastructure Layer
│   ├── aiService.js           # Prompt engineering & Groq response normalization
│   ├── deploymentService.js   # Project packaging & deployment status simulator
│   ├── groqClient.js          # Groq SDK client wrapper & JSON parser
│   ├── inviteService.js       # Resend API integration for email invitations
│   ├── platformUtils.js       # Helper functions (path normalization, version snapshotting)
│   ├── runtimeService.js      # Isolated temp file writing & child process spawning
│   └── socketServer.js        # Socket.io server, authentication, & presence state
├── app/                       # Next.js App Router Pages
│   ├── globals.css            # Tailwind CSS & global layout styles
│   ├── layout.tsx             # Root HTML wrapper and font loading
│   └── page.tsx               # Entry point rendering DevWatchApp component
├── components/                # Frontend React Components
│   ├── auth-modal.tsx         # Firebase Sign-In / Sign-Up modal
│   ├── cloud-ide.tsx          # Full Cloud IDE interface (Monaco, Terminals, AI drawer)
│   ├── devwatch-app.tsx       # Top-level view router (Landing vs Overview vs IDE)
│   ├── file-explorer.tsx      # Recursive file tree navigator (Create, Rename, Delete)
│   ├── landing-page.tsx       # Hero marketing page with feature showcase
│   ├── onboarding-flow.tsx    # New user onboarding modal
│   ├── ui.tsx                 # Reusable UI elements (Button, Input, Card, Modal)
│   └── workspace-shell.tsx    # Dashboard layout (Projects list, Activity feed, Team management)
├── lib/                       # Client-side Utilities & API Clients
│   ├── api.ts                 # Fetch API wrapper with automatic Auth token injection
│   ├── firebase.ts            # Client Firebase Web SDK setup
│   ├── types.ts               # TypeScript interfaces for all system entities
│   └── utils.ts               # Class merging (cn) and string utility helpers
├── server/                    # Secondary TypeScript Server Implementation (Zod & Refactor)
│   ├── index.ts               # Express TS server entry point (`npm run server:ts:dev`)
│   ├── routes.ts              # Zod-validated TypeScript API routes
│   ├── models.ts              # Mongoose models for TypeScript server
│   ├── store.ts               # In-memory mock store fallback when MongoDB is disabled
│   └── services/              # TS service counterparts
├── scripts/                   # Maintenance & Verification Scripts
│   └── verify-ai-persistence.js# DB seed & AI review persistence sanity check script
├── server.js                  # Primary Backend Entry Point (`npm start`, `npm run dev`)
└── package.json               # Project manifest & script runner
```

Dual-Server Architecture Analysis

An interviewer may notice both `server.js` and `server/index.ts` in the codebase.

- **Primary Production Server (`server.js`)**: Runs via `npm start` or `npm run dev`. Built using Node.js CommonJS (`require`). Handles full MongoDB persistence, Socket.io presence, Groq AI reviews, and process execution.
- **Refactoring Prototype (`server/index.ts`)**: Runs via `npm run server:ts:dev` using `tsx`. Uses TypeScript ESM imports, Zod schema validation (`loginSchema` , `createProjectSchema`), and an in-memory fallback store (server/store.ts) when MongoDB connection is omitted.

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4. REST API DOCUMENTATION & CONTRACTS

API Endpoint Specification Table

Method	Endpoint Route	Purpose	Middleware	Auth Required	Key Request Parameters / Body	Key Response Fields
`POST`	`/api/auth/login`	Authenticate user via Firebase token or credentials	`asyncHandler`	No	`{ idToken, uid, email, name, photoURL }`	`{ success: true, token, user, onboardingRequired }`
`POST`	`/api/auth/signup`	Register new user account	`asyncHandler`	No	`{ uid, email, name, photoURL }`	`{ success: true, token, user, onboardingRequired }`
`POST`	`/api/auth/onboarding/complete`	Mark user onboarding complete	`requireAuth`	Yes	None	`{ success: true, user }`
`GET`	`/api/workspaces`	List accessible workspaces for user	`requireAuth`	Yes	Query: None	`{ success: true, data: Workspace[] }`
`POST`	`/api/workspaces`	Create a new workspace	`requireAuth`	Yes	`{ workspaceName }`	`{ success: true, data: Workspace }`
`PATCH`	`/api/workspaces/:workspaceId`	Update workspace settings	`requireAuth`	Yes	`{ workspaceName, settings }`	`{ success: true, data: Workspace }`
`DELETE`	`/api/workspaces/:workspaceId`	Delete workspace & all child projects	`requireAuth`	Yes (Owner)	Params: `workspaceId`	`{ success: true, data: Workspace }`
`GET`	`/api/projects`	Fetch all projects in workspace	`requireAuth`	Yes	Query: `workspaceId`	`{ success: true, data: Project[] }`
`POST`	`/api/projects`	Create project in workspace	`requireAuth`	Yes (Dev+)	`{ workspaceId, projectName, description }`	`{ success: true, data: Project }`
`GET`	`/api/projects/:projectId`	Fetch project details	`requireAuth`	Yes	Params: `projectId`	`{ success: true, data: Project }`
`PATCH`	`/api/projects/:projectId`	Update project metadata	`requireAuth`	Yes	`{ projectName, description, status }`	`{ success: true, data: Project }`
`DELETE`	`/api/projects/:projectId`	Delete project & all associated files	`requireAuth`	Yes (Admin+)	Params: `projectId`	`{ success: true, data: Project }`
`GET`	`/api/projects/:projectId/export`	Download project as ZIP stream	`requireAuth`	Yes	Params: `projectId`	Binary ZIP stream (`Content-Type: application/zip`)
`GET`	`/api/files`	Get virtual file tree for project	`requireAuth`	Yes	Query: `projectId`	`{ success: true, data: File[] }`

Method	Endpoint Route	Purpose	Middleware	Auth Required	Key Request Parameters / Body	Key Response Fields
`POST`	`/api/files`	Create a new file or folder	`requireAuth`	Yes (Dev+)	`{ projectId, path, content, type, language }`	`{ success: true, data: File }`
`PATCH`	`/api/files/:fileId`	Rename, move, or edit file content	`requireAuth`	Yes (Dev+)	`{ path, content, language, expectedRevision }`	`{ success: true, data: File }`
`DELETE`	`/api/files/:fileId`	Delete file or folder recursively	`requireAuth`	Yes (Dev+)	Params: `fileId`	`{ success: true, data: { deleted: File[] } }`
`POST`	`/api/files/import`	Bulk import files into project	`requireAuth`	Yes (Dev+)	`{ projectId, files: [{ path, content }] }`	`{ success: true, data: { created, rejected } }`
`POST`	`/api/files/:fileId/duplicate`	Duplicate a file or folder	`requireAuth`	Yes (Dev+)	`{ path }`	`{ success: true, data: File[] }`
`GET`	`/api/files/:fileId/versions`	List revision history of a file	`requireAuth`	Yes	Params: `fileId`	`{ success: true, data: Version[] }`
`POST`	`/api/versions/:versionId/restore`	Restore file to a previous version	`requireAuth`	Yes (Dev+)	Params: `versionId`	`{ success: true, data: File }`
`POST`	`/api/ai/review`	Run AI code review on file/project	`requireAuth`	Yes	`{ projectId, fileId, code, analysisType }`	`{ success: true, type: 'code_review', data: Review }`
`POST`	`/api/ai/suggestions`	Request code improvement suggestions	`requireAuth`	Yes	`{ filePath, code, context }`	`{ success: true, type: 'code_suggestions', data }`
`POST`	`/api/ai/bug-detection`	Detect potential bugs and edge cases	`requireAuth`	Yes	`{ filePath, code, context }`	`{ success: true, type: 'bug_detection', data }`
`POST`	`/api/ai/auto-fix`	Generate automated bug fixes	`requireAuth`	Yes	`{ filePath, code, issue }`	`{ success: true, type: 'auto_fix', data }`
`POST`	`/api/ai/chat`	AI coding assistant context Q&A	`requireAuth`	Yes	`{ projectId, question, context }`	`{ success: true, type: 'chat_assistant', data }`
`POST`	`/api/ai/project-analysis`	Full multi-file repository evaluation	`requireAuth`	Yes	`{ projectId }`	`{ success: true, type: 'project_analysis', data }`
`POST`	`/api/projects/:projectId/run`	Execute project code in temp sandbox	`requireAuth`	Yes (Dev+)	Params: `projectId`	`{ success: true, data: Execution }`

Method	Endpoint Route	Purpose	Middleware	Auth Required	Key Request Parameters / Body	Key Response Fields
`POST`	`/api/executions/:executionId/stop`	Kill running subprocess sandbox	`requireAuth`	Yes (Dev+)	Params: `executionId`	`{ success: true, data: Execution }`
`POST`	`/api/projects/:projectId/deploy`	Generate build & simulation deployment	`requireAuth`	Yes (Dev+)	Params: `projectId`, Body: `{ environment }`	`{ success: true, data: Deployment }`
`POST`	`/api/invites`	Invite member via email	`requireAuth`	Yes (Admin+)	`{ workspaceId, email, role, projectId }`	`{ success: true, data: Invitation }`
`POST`	`/api/invites/:token/accept`	Accept workspace invitation	`requireAuth`	Yes	Params: `token`	`{ success: true, data: Member }`
`POST`	`/api/invites/:token/reject`	Reject workspace invitation	`requireAuth`	Yes	Params: `token`	`{ success: true, data: Invitation }`
`GET`	`/api/dashboard`	Fetch consolidated workspace telemetry	`requireAuth`	Yes	Query: `workspaceId`	`{ success: true, data: DashboardMetrics }`
`GET`	`/api/search`	Search files & projects in workspace	`requireAuth`	Yes	Query: `workspaceId, q`	`{ success: true, data: { projects, files } }`

5. AUTHENTICATION, SECURITY & RBAC DEEP DIVE

Authentication & Token Flow



1. **Client Identity Verification**: Users authenticate on the frontend via Firebase Web Auth SDK.

2. **Backend Authentication Exchange**: The frontend sends the Firebase ID token to POST `/api/auth/login`.
3. **Token Verification**: firebaseAdmin.js invokes `admin.auth().verifyIdToken(idToken)`.
4. **User Synchronization**: `upsertUser` locates or creates the MongoDB `User` document by `firebaseUid` or `email`.
5. **Session Signing**: auth.js signs a 7-day JWT containing `{ userId, uid, email }` using `process.env.JWT_SECRET`.
6. **Protected API Access**: Client passes the JWT in the `Authorization: Bearer <token>` header for all subsequent API and Socket.io requests.

Role-Based Access Control (RBAC) Matrix

DevWatchAI implements granular workspace-level access control enforced via `getWorkspaceMembership()` in platformRoutes.js:

Role	Workspace Settings	Invite / Remove Members	Create / Edit Projects	Edit / Delete Files	Run Execution	Trigger Deployments	View Workspace
Owner	✅ Full	✅ Full	✅ Full	✅ Full	✅ Full	✅ Full	✅ Full
Admin	❌ No	✅ Full	✅ Full	✅ Full	✅ Full	✅ Full	✅ Full
Developer	❌ No	❌ No	✅ Create/Edit	✅ Full	✅ Full	✅ Full	✅ Full
Viewer	❌ No	❌ No	❌ Read Only	❌ Read Only	❌ Read Only	❌ Read Only	✅ Full

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Security Safeguards

1. Path Traversal & File Injection Defense

In runtimeService.js, `safeRelativePath()` protects against directory traversal attacks (e.g., `../../etc/passwd`):

```
function safeRelativePath(value) {
  const normalized = String(value || "").replace(/\\/g, "/").replace(/^\/+/, "");
  const segments = normalized.split("/").filter(Boolean);
  if (!segments.length || segments.some((segment) => segment === "..")) {
    throw new Error("Unsafe project file path.");
  }
  return segments.join(path.sep);
}
```

2. Command Injection Defense

Process execution in `runtimeService.js` uses `child_process.spawn` with `shell: false`. This passes parameters directly to the operating system execution array without invoking a shell binary (`/bin/sh` or `cmd.exe`), nullifying shell-string injection vectors like `; rm -rf /`.

3. NoSQL Injection Defense

Search and filtering endpoints wrap raw input strings with `escapeRegex()` before passing them into MongoDB `$regex` queries:

```
function escapeRegex(value) {
  return String(value).replace(/[.*+?^${}()|[\]\\]/g, "\\$&");
}
```

Senior Interview Security Q&A

Q1: How does DevWatchAI handle token verification if Firebase Admin is unconfigured in a local environment?

> *Answer*: In `platformRoutes.js`, if `isFirebaseAdminConfigured()` returns `false` or token verification fails in non-production environments (`process.env.NODE_ENV !== 'production'`), the server falls back to parsing the mock user payload passed in the request body for local dev convenience. In production, however, missing or invalid Firebase ID tokens explicitly throw a `401 Unauthorized` error.

Q2: What is a major security vulnerability in the current code execution engine?

> *Answer*: Subprocesses spawned by `runtimeService.js` run under the host Node process's user privileges without OS container isolation (like Docker containers, microVMs such as AWS Firecracker, or gVisor). A malicious project script could inspect host system environment variables (`process.env`), execute network scans, or launch fork bombs. In a production environment, code execution must be dispatched to isolated, ephemeral Docker containers with read-only root filesystems, strict cgroup memory/CPU limits, and network namespace restrictions.

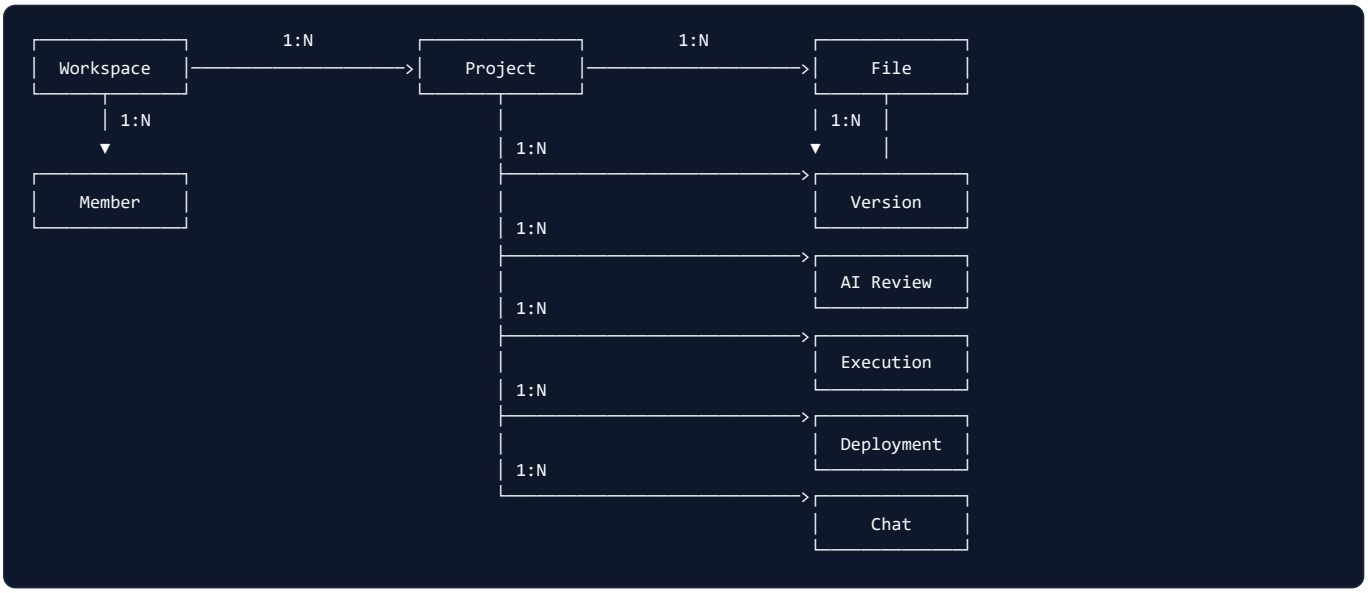
Q3: How does the application prevent optimistic lock race conditions during concurrent file saves?

> *Answer*: In `extendedRoutes.js` (`PATCH /api/files/:fileId`), the endpoint supports an `expectedRevision` check:
>
> ```
> if (Number.isInteger(request.body.expectedRevision) && request.body.expectedRevision !== file.revision) {
>   return response.status(409).json({ success: false, error: "This file changed on another client.", code: "REVISION_CONFLICT"
> }
> ```
>
> If another user updated the file in the interim, `file.revision` increments, causing out-of-date save requests to fail with a `409 Conflict` status code.

---

6. DATABASE DEEP ANALYSIS & DATA MODELS

Entity Relationship Mapping



Schema & Index Audit

1. File Collection (models/File.js)

- **Fields**: `projectId` (ObjectId), `parentId` (ObjectId), `name` (String), `path` (String), `content` (String), `type` ('file' | 'folder'), `language` (String), `size` (Number), `revision` (Number), `lastEditedBy` (ObjectId).
- **Index**: `{ projectId: 1, path: 1 }` (Unique).
- **Optimization Audit**: The text search endpoint (`/api/search`) attempts `$text: { $search: query }`. However, `File.js` lacks an explicit text index definition on `content` and `path`. As a result, text searches trigger a fallback `$regex` scan. Adding `FileSchema.index({ name: "text", path: "text", content: "text" })` is necessary for high-performance full-text code search.

2. Member Collection (models/Member.js)

- **Fields**: `workspaceId` (ObjectId), `userId` (ObjectId), `role` ('owner'|'admin'|'developer'|'viewer'), `status` ('active'|'invited'|'removed').
- **Index**: `{ workspaceId: 1, userId: 1 }` (Unique).

3. Execution Collection (models/Execution.js)

- **Fields**: `workspaceId`, `projectId`, `triggeredBy`, `runtime`, `entryFile`, `status` ('queued'|'running'|'success'|'failed'|'timed-out'|'stopped'), `exitCode`, `stdout`, `stderr`, `durationMs`.
- **Index**: `{ projectId: 1, createdAt: -1 }`.

---

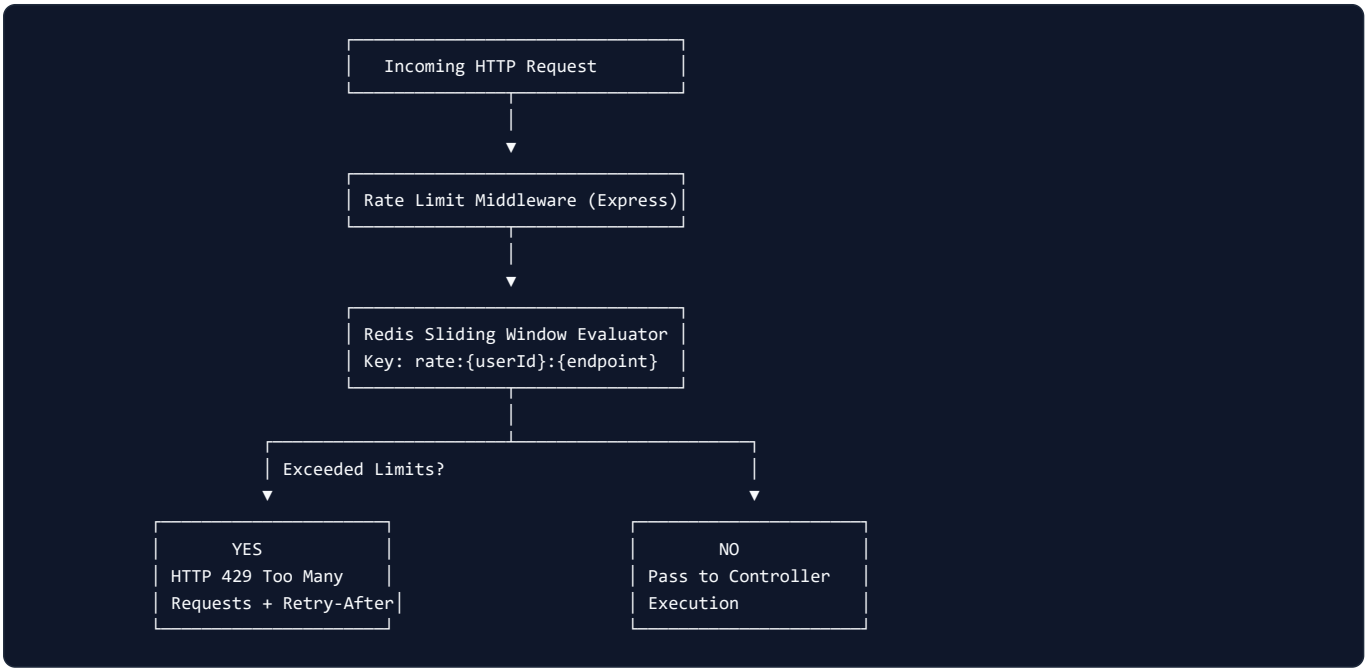
7. API PERFORMANCE, RATE LIMITING & DDoS PROTECTION

Rate Limiting Audit (Current vs Proposed Architecture)

- **Current State**: The repository currently **does not have rate-limiting middleware** configured in `server.js`.
- **Vulnerability Risk**:
  1. `/api/ai/*`: High financial cost exposure via Groq API token consumption abuse.
  2. `/api/projects/:projectId/run`: Vulnerable to CPU starvation caused by continuous process spawning.
  3. `/api/invites`: Vulnerable to transactional email spamming via Resend.

Production Rate Limiting Implementation Architecture

To mitigate these risks, a production implementation should use a **Redis-backed Sliding Window Counter** rate limiter:



Proposed Middleware Code Blueprint (`middleware/rateLimiter.js`)

```
const rateLimit = require("express-rate-limit");
const RedisStore = require("rate-limit-redis");

const aiRateLimiter = rateLimit({
  windowMs: 15 * 60 * 1000, // 15 minutes
  max: 30, // Limit each user to 30 AI requests per window
  standardHeaders: true,
  legacyHeaders: false,
  keyGenerator: (req) => req.user?._id || req.ip,
  message: { success: false, error: "AI rate limit exceeded. Please wait 15 minutes." }
});

const executionRateLimiter = rateLimit({
  windowMs: 1 * 60 * 1000, // 1 minute
  max: 5, // Max 5 code executions per minute
  keyGenerator: (req) => req.user?._id || req.ip,
  message: { success: false, error: "Execution quota exceeded. Try again in 1 minute." }
});

module.exports = { aiRateLimiter, executionRateLimiter };
```

---

8. ERROR HANDLING & RESILIENCE ARCHITECTURE

Current Strategy Analysis

1. **Controller Async Wrapper**: All route handlers are wrapped in `asyncHandler(fn)` in platformRoutes.js, automatically catching rejected Promises and forwarding errors to Express `next(error)` :

   ```
   function asyncHandler(handler) {
     return (request, response, next) => {
       Promise.resolve(handler(request, response, next)).catch(next);
     };
   }
   ```

2. **Centralized Error Middleware**: In `server.js` (lines 64-69):

   ```
   app.use((error, _request, response, _next) => {
     console.error(error);
     response.status(error.status || 500).json({
       message: error.message || "Internal server error"
     });
   });
   ```

3. **Frontend Error Normalization**: In lib/api.ts, non-2xx responses throw an `ApiError` instance containing `status` , `message` , and raw server `payload` .

---

9. SENIOR BACKEND DESIGN INTERVIEW QUESTIONS & ANSWERS

Q1: Why did you choose Express and MongoDB over a relational database like PostgreSQL for DevWatchAI?

Answer:

> "DevWatchAI's primary entity is the project file tree. Files, folders, nested AI code reviews, execution logs, and activity metadata have highly dynamic, schema-less properties. MongoDB's document model allows us to store complete file nodes and version snapshots natively as JSON-like BSON documents without requiring complex multi-table JOINs across directories, revisions, and metadata. Additionally, MongoDB's change streams and high write throughput fit real-time log ingestion and version snapshotting.
> However, if strict ACID transaction guarantees were required across workspace billing, multi-user role assignments, and subscription limits, PostgreSQL with JSONB columns would be an equally strong candidate."

Q2: How would you scale DevWatchAI to support 1 million concurrent active users?

Answer:

> "Scaling DevWatchAI to 1 million users requires decoupling stateless API serving from stateful WebSocket presence and process execution:
> 1. **Stateless Express API Nodes**: Deploy HTTP API servers behind an AWS ALB or Cloudflare Load Balancer across an auto-scaling Kubernetes cluster (EKS).
> 2. **Distributed WebSocket Architecture**: Dedicated Socket.io servers scaled horizontally using `@socket.io/redis-adapter`. This guarantees that WebSocket events emitted on Server A reach clients connected to Server B.
> 3. **Isolated Remote Code Execution Fleet**: Decouple `runtimeService.js` from the main web server. Offload code execution requests via a message queue (AWS SQS or RabbitMQ) to a pool of isolated worker instances running microVMs (Firecracker or gVisor-sandboxed Docker containers).
> 4. **Database Sharding**: Shard MongoDB clusters on `workspaceId` to ensure all data for a workspace resides on the same primary shard, minimizing cross-shard latency."

Q3: How would you handle real-time co-editing conflicts if two users edit the same file simultaneously?

Answer:

> "Currently, DevWatchAI relies on optimistic revision lock detection (`expectedRevision` in `PATCH /api/files/:fileId`) and WebSocket `file:change` broadcasting. To support true simultaneous multi-user collaborative text editing (like Google Docs or Figma), we would replace naive revision checks with **Conflict-free Replicated Data Types (CRDTs)** such as **Yjs** or **Automerge**, integrated directly with the Monaco Editor.
> Client keystrokes would produce binary CRDT state vectors transmitted over Socket.io. The CRDT algorithm resolves non-sequential edits deterministically across all peers without requiring central server lock step."

---

10. FRONTEND INTERVIEW QUESTIONS & ANSWERS

Q1: How is the Monaco Editor integrated into Next.js to prevent hydration mismatches?

Answer:

> "Monaco Editor relies on DOM canvas measurement and browser worker APIs which fail when evaluated during Server-Side Rendering (SSR). In cloud-ide.tsx, we dynamically import `@monaco-editor/react` with `{ ssr: false }`. We display a lightweight CSS loader fallback during initial client mount."

Q2: How does the IDE manage state transitions between files, WebSockets, and terminal outputs?

Answer:

> "The IDE maintains a unified state model in `cloud-ide.tsx`. Active file selection updates `activeFileId`. Changes inside Monaco trigger a debounced save handler. Simultaneously, the component mounts a `useEffect` hook that registers Socket.io listeners (`file:change`, `execution:log`, `presence:list`). When `execution:log` fires, the log string appends to `consoleLogs` state, driving real-time terminal renders."

---

11. DEVOPS, DEPLOYMENT & OBSERVABILITY

Production Dockerfile Blueprint (`Dockerfile`)

```
# Build Stage
FROM node:20-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npm run build

# Production Stage
FROM node:20-alpine AS runner
WORKDIR /app
ENV NODE_ENV=production
COPY package*.json ./
RUN npm ci --only=production
COPY --from=builder /app/.next ./.next
COPY --from=builder /app/public ./public
COPY --from=builder /app/server.js ./server.js
COPY --from=builder /app/routes ./routes
COPY --from=builder /app/services ./services
COPY --from=builder /app/models ./models
COPY --from=builder /app/config ./config
COPY --from=builder /app/middleware ./middleware

EXPOSE 4000
CMD ["node", "server.js"]
```

Key Environment Variables Checklist (`.env.example`)

```
PORT=4000
CLIENT_ORIGIN=http://localhost:3000
MONGODB_URI=mongodb://localhost:27017/devwatch
MONGODB_MAX_POOL_SIZE=10
JWT_SECRET=your-super-secret-jwt-key
GROQ_API_KEY=gsk_your_groq_api_key_here
FIREBASE_PROJECT_ID=devwatch-ai
FIREBASE_CLIENT_EMAIL=firebase-adminsdk@devwatch-ai.iam.gserviceaccount.com
FIREBASE_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\n..."
RESEND_API_KEY=re_your_resend_key
EMAIL_FROM=notifications@devwatch.ai
```

---

12. CODE QUALITY & ARCHITECTURAL REVIEW (SENIOR AUDIT)

Findings & Remediation Plan

1. Technical Debt: Dual Server Codebase Drift

- **Issue**: The project contains two parallel backend implementations: `server.js` (CommonJS) and `server/index.ts` (TypeScript + Zod).
- **Remediation**: Complete the migration to `server/index.ts`, establish unified Zod request validation across all endpoints, and remove `server.js` to eliminate code duplication.

2. Security Vulnerability: Hardcoded JWT Fallback Secret

- **Issue**: In `middleware/auth.js`, fallback secret `"devwatch-local-secret"` is used if `process.env.JWT_SECRET` is missing.
- **Remediation**: Throw a startup fatal error in `server.js` if `process.env.JWT_SECRET` is undefined in production.

3. System Stability: Unbounded Code Execution Concurrency

- **Issue**: `runtimeService.js` spawns subprocesses directly on the host machine without concurrency caps.
- **Remediation**: Implement a worker queue (e.g., `bullmq` with Redis) limiting concurrent executions to `N` processes matching CPU core count.

---

13. INTERVIEW CHEAT SHEET & QUICK REFERENCE

A) 2-Minute Elevator Pitch

> "DevWatchAI is an AI-native, collaborative cloud IDE and engineering workspace built with Next.js, Express, MongoDB, Socket.io, and Groq LLMs. It solves the friction of fragmented software development by unifying Monaco-based cloud editing, Google Docs-style real-time collaboration, instant subprocess code execution, and high-speed AI code reviews into a single browser environment.
> On the backend, we use Express and Mongoose for document persistence, Socket.io for low-latency cursor presence and log streaming, and custom process runners that isolate user code in temporary OS sandboxes. Our AI features run on Groq's `llama-3.3-70b-versatile` model, using strict JSON mode schemas to deliver instant code scores, bug detections, and auto-fixes."

B) 5-Minute Technical Deep-Dive Narrative

> "To understand DevWatchAI's architecture, we look at three core layers:
> First, **State & Workspace Management**: We represent projects as virtual filesystem trees in MongoDB (`File` collection), where files and folders exist as parent-child nodes indexed on `projectId` and `path`. File mutations automatically generate snapshot records in our `Version` collection, allowing instant point-in-time rollbacks.
> Second, **Real-Time Collaboration & Execution**: We attach Socket.io to our Node HTTP server. When a developer opens a project, they join a project-specific room. Our in-memory presence map tracks active cursor positions, open files, and typing states across peers. For code execution, when a user hits 'Run', `runtimeService.js` provisions an isolated OS temporary directory via `fs.mkdtemp`, writes the project files from MongoDB, auto-detects entry points (`package.json` , `index.ts` , `main.py`), and spawns a non-shell child process (`node` , `tsx` , `python`). We capture `stdout` and `stderr` streams, emitting them over WebSockets directly to the user's terminal UI.
> Third, **AI Pipeline & Security**: We interface with Groq's ultra-low-latency LPU platform. To guarantee structured JSON output without prose, we enforce strict JSON mode expectations in our system prompts and parse raw responses through validation routines. For security, we prevent directory traversal via `safeRelativePath` checks, enforce RBAC (Owner, Admin, Developer, Viewer) across endpoints, and sign 7-day JWT session tokens following initial Firebase ID token exchange."

C) 20 Most Likely Interviewer Questions & Concise Answers

1. **Q: How are files stored in MongoDB?**
   - *A*: As virtual nodes in the `File` collection containing `path` , `content` , `type` ('file'|'folder'), `parentId` , and `revision` .
2. **Q: What LLM model is used and why?**
   - *A*: Groq's `llama-3.3-70b-versatile` due to its high token throughput (>500 tokens/sec) and structured JSON mode support.
3. **Q: How does authentication work?**
   - *A*: Client authenticates via Firebase Web SDK -> sends ID token to POST `/api/auth/login` -> Server verifies ID token via Firebase Admin -> signs and returns a 7-day custom JWT session token.
4. **Q: How does real-time presence work?**
   - *A*: Socket.io room membership (`project:projectId`) tracks socket IDs and cursor positions in an in-memory `projectPresence` Map and broadcasts `presence:list` updates.
5. **Q: How is code executed in the browser?**

- *A*: Files are written to a host temp directory via `fs.mkdtemp`, and a child process (`node` / `python` / `tsx`) is spawned with stdout/stderr piped over WebSockets.

6. **Q: How do you prevent directory traversal attacks during execution?**
   - *A*: Paths are validated with `safeRelativePath()`, which normalizes paths and rejects any path containing `..` or unsafe path segments.
7. **Q: Why are Monaco Editor imports dynamic in Next.js?**
   - *A*: Monaco depends on browser DOM APIs (`window`, `document`). Dynamic importing with `{ ssr: false }` prevents SSR hydration errors.
8. **Q: What happens if two users save the same file at the same time?**
   - *A*: Optimistic locking check (`expectedRevision === file.revision`). If revisions conflict, server responds with `409 Conflict`.
9. **Q: How does project export work?**
   - *A*: GET `/api/projects/:projectId/export` queries all project files from MongoDB and pipes a zip stream directly to the response using `archiver`.
10. **Q: What database indexes exist on the File collection?**
    - *A*: Compound unique index on `{ projectId: 1, path: 1 }`.
11. **Q: How are invitations sent to new workspace members?**
    - *A*: Server generates a 32-byte random token, stores its `sha256` hash in MongoDB `Invitation`, and emails an invite link via Resend API.
12. **Q: What RBAC roles exist?**
    - *A*: Owner, Admin, Developer, Viewer.
13. **Q: How are TypeScript files executed?**
    - *A*: Spawns Node.js targeting `tsx/cli` as the executor binary.
14. **Q: What is the maximum execution timeout?**
    - *A*: 30 seconds (`DEFAULT_TIMEOUT_MS = 30000`), after which `child.kill()` terminates the process.
15. **Q: How are AI scores calculated?**
    - *A*: Groq returns scores for quality, maintainability, security, and performance (0-100), normalized via `normalizeScore()`.
16. **Q: How does full project search work?**
    - *A*: Regex matching on project name/description and fallback regex matching on file path and content.
17. **Q: What rate limiting is implemented?**
    - *A*: Currently missing; proposed architecture uses Redis sliding window counters (`express-rate-limit` + Redis store).
18. **Q: What is the difference between `server.js` and `server/index.ts`?**
    - *A*: `server.js` is the CommonJS production server; `server/index.ts` is a TypeScript refactor using Zod validation schemas.
19. **Q: How are error lines parsed during execution?**
    - *A*: Regex matching on stderr output (`parseErrorLocation`) extracts file path and line number to highlight in the IDE editor.
20. **Q: How would you scale WebSockets horizontally?**
    - *A*: Attach `@socket.io/redis-adapter` to publish/subscribe socket events across a cluster of Socket.io nodes.

D) "Things Interviewers May Catch Me On" & Immediate Defenses

1. **Host Subprocess Execution**: "Currently code runs via Node's `child_process.spawn` on the server host. In production, I would migrate process execution to isolated ephemeral Docker/Firecracker containers."
2. **Missing Rate Limiting**: "The current MVP relies on frontend throttling. The next item on our roadmap is adding Redis-backed rate limiting middleware for `/api/ai/*` and `/api/projects/:projectId/run`."
3. **Missing MongoDB Text Index**: "Full-text file search uses regex fallback because explicit `$text` indexes aren't defined on `FileSchema`. Adding a compound text index on `path` and `content` is a straightforward index migration."

14. COMPLETE INTERVIEW QUESTION BANK (CATEGORIES A - N)

A) PROJECT UNDERSTANDING QUESTIONS

Q1: What core business or developer pain point does DevWatchAI address?

- **Difficulty Level**: Easy
- **Why Interviewer Asks**: Tests if you understand the product vision beyond just writing syntax.
- **Ideal Answer**: Standard developer workflows suffer from context switching across disconnected tools: local IDEs, external pull-request review portals, separate terminal sandboxes, and monitoring dashboards. DevWatchAI consolidates cloud editing (Monaco), real-time pair-programming collaboration (Socket.io), automated AI code auditing & fixing (Groq LLM), isolated code execution, and telemetry into a single unified browser workspace.
- **Code Reference**: components/landing-page.tsx & components/cloud-ide.tsx

Q2: How does a user transition from signup to creating and running their first project?

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Evaluates end-to-end knowledge of user onboarding and data hierarchy.
- **Ideal Answer**:
  1. User authenticates via Firebase Auth in components/auth-modal.tsx.
  2. Server verifies Firebase token and returns a JWT session via `POST /api/auth/signup` in routes/platformRoutes.js.
  3. If `onboardingRequired` is true, components/onboarding-flow.tsx guides workspace setup.
  4. `POST /api/workspaces` creates the workspace and sets the user as `owner` in `Member` collection.
  5. `POST /api/projects` initializes the project container.
  6. `POST /api/files` populates default boilerplate files (`index.ts` , `package.json`).
  7. User clicks "Run Code", triggering `POST /api/projects/:projectId/run` .
- **Code Reference**: components/devwatch-app.tsx

Q3: What happens when an invited member accepts a workspace invitation?

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Checks understanding of invitation lifecycle, token hashing, and authorization propagation.
- **Ideal Answer**: The invitation URL contains a raw hex token. When the recipient clicks Accept, `POST /api/invites/:token/accept` hashes the token using `sha256` in services/platformUtils.js and searches `Invitation` collection for a matching `pending` record. Upon verification, a new `Member` document is upserted with the designated role (`admin` , `developer` , `viewer`), the project's member list is updated, and invitation status transitions to `accepted` .
- **Code Reference**: routes/extendedRoutes.js

B) ARCHITECTURE QUESTIONS

Q4: Why is the backend structured with Express & Node.js rather than a serverless architecture like Next.js API routes or AWS Lambda?

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Tests architectural decision-making around persistent state, WebSockets, and long-running subprocesses.
- **Ideal Answer**: While Next.js handles frontend page rendering, serverless functions (like Vercel serverless or AWS Lambda) are stateless and time-bound (15-second timeout execution limits). DevWatchAI requires:
  1. **Persistent WebSocket Connections**: Socket.io server in services/socketServer.js maintains active room connections for live presences and log streaming.
  2. **Subprocess Lifecycle Control**: `runtimeService.js` spawns and monitors long-running Node/Python binaries via `child_process.spawn` . Serverless functions cannot maintain child process handles across HTTP requests.
- **Code Reference**: server.js

Q5: Explain the architectural trade-offs of storing project code in MongoDB vs. a host filesystem or S3 bucket.

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Evaluates knowledge of storage trade-offs, concurrency, and I/O bottlenecks.
- **Ideal Answer**:
  - **MongoDB Storage (Current)**:

    - *Pros*: Atomic document updates, single source of truth, easy version snapshotting, simplified transactional rollbacks, and effortless horizontal replication without mounting network file shares (NFS).
    - *Cons*: High memory footprint when serving large binary files; file sizes limited to 16MB GridFS limits if unchunked.
  - **Host Filesystem / S3 Storage (Alternative)**:
    - *Pros*: Native POSIX file system performance for local compilers.
    - *Cons*: Storing files on host disk makes API servers stateful, breaking horizontal auto-scaling unless using AWS EFS.
- **Code Reference**: models/File.js & services/runtimeService.js

Q6: How does the dual-server setup (`server.js` vs `server/index.ts`) work, and how would you resolve it?

- **Difficulty Level**: Senior / Expert
- **Why Interviewer Asks**: Tests code audit ability and refactoring strategy.
- **Ideal Answer**: `server.js` is the CommonJS production server loaded by default (`npm start`), relying on standard Express routes. `server/index.ts` is a modern TypeScript refactor utilizing ESM syntax and Zod input validation schemas. However, having both creates architectural drift. My refactoring roadmap is to complete Zod validation parity on `server/index.ts` , port Socket.io event typings, update `package.json` script targets to `server/index.ts` , and deprecate `server.js` .
- **Code Reference**: server.js & server/index.ts

C) FRONTEND QUESTIONS

Q7: How does Monaco Editor communicate code edits back to the backend without lagging the UI?

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Tests knowledge of React rendering, event debouncing, and optimistic state updates.
- **Ideal Answer**: In components/cloud-ide.tsx, Monaco's `onChange` event updates local React state immediately for fluid 60fps typing. To prevent flooding the API with HTTP requests on every single keypress, saving is debounced. Additionally, real-time typing indicators are emitted over Socket.io via `file:change` and `presence:file` events, notifying co-editors without writing uncommitted edits to MongoDB.
- **Code Reference**: components/cloud-ide.tsx

Q8: What is the purpose of `customizations` and `useMemo` hooks in the Cloud IDE layout?

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Evaluates React performance optimization techniques.
- **Ideal Answer**: In `cloud-ide.tsx` , `useMemo` memoizes heavy computations like sorting directory trees, calculating active presence counts, and parsing execution logs. This avoids costly tree re-computations whenever un-related state changes occur (such as modal toggles or chat message keystrokes).
- **Code Reference**: components/cloud-ide.tsx

Q9: How does the client maintain persistent authentication across page reloads?

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Tests browser storage handling and session hydration.
- **Ideal Answer**: Upon successful sign-in via `AuthModal` , the JWT token and user profile object are serialized to browser `localStorage` under `devwatch_session` in components/devwatch-app.tsx. On initial component mount (`useEffect`), `DevWatchApp` reads `localStorage` , validates token existence, and hydrates user state. When calling `apiFetch` , if no explicit token is passed, it automatically retrieves fresh ID tokens via `auth.currentUser.getIdToken()` .
- **Code Reference**: lib/api.ts

D) BACKEND QUESTIONS

Q10: Walk through the complete lifecycle of an incoming HTTP request in Express.

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Tests fundamental understanding of Node middleware chains.
- **Ideal Answer**:

1. Request arrives at Node `http.Server` on port 4000.
2. CORS middleware checks origin against whitelist.
3. Body parser middlewares (`express.json({ limit: '10mb' })`) parse JSON payloads.
4. `requireAuth` middleware extracts Bearer token, verifies JWT with `jwt.verify`, queries `User` document in MongoDB, and attaches `request.user`.
5. Express matches route (e.g. `POST /api/files`).
6. Route executes inside `asyncHandler`.
7. If an error is thrown, `asyncHandler` catches it and calls `next(error)`.
8. Centralized error middleware formats status code and JSON payload `{ message: error.message }`.

- **Code Reference**: server.js & middleware/auth.js

Q11: How does `runtimeService.js` safely spawn and manage code execution subprocesses?

- **Difficulty Level**: Hard / Senior
- **Why Interviewer Asks**: Assesses operating system integration, stream piping, and timeout enforcement.
- **Ideal Answer**:
  1. `startExecution` creates a unique temp folder on OS disk via `fs.mkdtemp(path.join(os.tmpdir(), 'devwatch-run-'))`.
  2. Iterates over `files` array from MongoDB and writes them to temp disk via `fs.writeFile`.
  3. `selectRunner` inspects file extensions/package.json to select executable binary (`node` , `tsx` , or `python`).
  4. Spawns process using `child_process.spawn(command, args, { cwd, shell: false })`.
  5. Listens to `stdout.on('data')` and `stderr.on('data')`, appending output buffer and emitting `execution:log` over Socket.io.
  6. Enforces a 30-second timeout (`setTimeout`) which invokes `child.kill()` if process exceeds quota.
  7. On process `close` or `error`, `finalizeExecution` records duration, updates `Execution` and `Monitoring` MongoDB schemas, and removes temp directory via `fs.rm`.
- **Code Reference**: services/runtimeService.js

Q12: How is Groq API integrated for low-latency AI Code Reviews?

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Evaluates AI LLM integration, prompt engineering, and structured JSON parsing.
- **Ideal Answer**: `aiService.js` constructs a system prompt mandating production engineering analysis and forcing JSON output. It calls `createGroqChatCompletion` in services/groqClient.js using model `llama-3.3-70b-versatile` with `response_format: { type: "json_object" }`. The response is parsed via `extractJson()`, normalized into scores (0-100) and issue objects, saved in MongoDB `AIReview` collection, and updates project status to `Reviewing` or `Warning`.
- **Code Reference**: services/aiService.js

E) DATABASE QUESTIONS

Q13: Explain the indexing strategy on the `File` collection and how it impacts performance.

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Evaluates MongoDB query optimization and unique constraints.
- **Ideal Answer**: In models/File.js, we define a compound unique index on `{ projectId: 1, path: 1 }`. This ensures:
  1. Rapid lookup of files within a project by exact path (`File.findOne({ projectId, path })`).
  2. Prevention of duplicate files or folders residing at the exact same directory path.
  3. Efficient regex prefix scans for directory folder deletes (`path: { $regex: '^/src/' }`).
- **Code Reference**: models/File.js

Q14: How does DevWatchAI handle versioning and historical rollbacks for files?

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Tests data modeling for historical version snapshots.

- **Ideal Answer**: Whenever a file action occurs (create, save, rename, move, restore, import, delete), `createVersion()` in services/platformUtils.js creates an immutable document in the `Version` collection storing `fileId`, `projectId`, snapshot `content`, `path`, `revision`, and `createdBy`. Restoring a version (`POST /api/versions/:versionId/restore`) fetches the target version snapshot and updates the active `File` document while incrementing `revision`.
- **Code Reference**: services/platformUtils.js & routes/extendedRoutes.js

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F) API QUESTIONS

Q15: Walk through `POST /api/projects/:projectId/export`. How does streaming work?

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Tests understanding of HTTP streaming, memory management, and zip archive generation.
- **Ideal Answer**: Rather than writing a zip archive to server disk and reading it back, the endpoint streams the archive directly to the client response:
  1. Sets response headers: `Content-Type: application/zip` and `Content-Disposition: attachment; filename="project.zip"`.
  2. Instantiates `archiver('zip', { zlib: { level: 9 } })`.
  3. Pipes archive directly to HTTP response: `archive.pipe(response)`.
  4. Queries project files from MongoDB and loops through each, appending file content via `archive.append(content, { name: zipPath })`.
  5. Finalizes archive via `archive.finalize()`.
- **Code Reference**: routes/platformRoutes.js

Q16: Why does `POST /api/files/:fileId` return a 409 status code during revision conflicts?

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Tests knowledge of HTTP status codes and concurrent editing guards.
- **Ideal Answer**: If the client provides an `expectedRevision` in the request body that does not match the stored `file.revision` in MongoDB, it indicates that another user modified the file in the interim. Returning `HTTP 409 Conflict` alerts the client to conflict state without silently overwriting changes.
- **Code Reference**: routes/extendedRoutes.js

G) SECURITY QUESTIONS

Q17: How does DevWatchAI prevent Directory Traversal attacks when running user code?

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Tests security awareness regarding file system access vulnerability.
- **Ideal Answer**: In services/runtimeService.js, `safeRelativePath()` strips leading slashes, normalizes backslashes, splits path segments, and explicitly checks if any segment equals `..`. If `..` is detected, an "Unsafe project file path" exception is thrown, preventing users from writing files outside the temporary execution directory (e.g. `../../etc/passwd`).
- **Code Reference**: services/runtimeService.js

Q18: What is the risk of using `child_process.spawn` vs `child_process.exec`?

- **Difficulty Level**: Senior / Expert
- **Why Interviewer Asks**: Tests deep knowledge of Node process execution security.
- **Ideal Answer**: `child_process.exec` passes command strings directly to a shell binary (`/bin/sh` or `cmd.exe`), making it highly vulnerable to Command Injection (e.g., passing `filename.js; rm -rf /`). `child_process.spawn` with `shell: false` executes the binary executable directly with an argument vector array, completely bypassing shell string evaluation and eliminating shell injection risks.
- **Code Reference**: services/runtimeService.js

H) SYSTEM DESIGN QUESTIONS BASED ON THIS PROJECT

Q19: How would you scale Socket.io presence across 50 server instances?

- **Difficulty Level**: Expert
- **Why Interviewer Asks**: Tests horizontal scaling of stateful WebSocket connections.
- **Ideal Answer**: By default, Socket.io rooms exist only in the memory of a single Node process. To scale horizontally across 50 instances behind a load balancer, we would attach `@socket.io/redis-adapter`. When a socket on Server 1 emits a presence update or log event to `project:123`, the Redis Adapter publishes the payload to a Redis Pub/Sub channel. All 50 Socket.io instances subscribe to Redis, broadcasting the event to locally connected clients seamlessly.
- **Code Reference**: services/socketServer.js

Q20: How would you isolate user code execution in production instead of running on the main host server?

- **Difficulty Level**: Expert
- **Why Interviewer Asks**: Assesses production sandbox design and container orchestration.
- **Ideal Answer**: Host execution in `runtimeService.js` is an MVP vulnerability. In production, we would decouple execution into a dedicated worker fleet:
  1. HTTP `/run` endpoint publishes an execution job to an AWS SQS queue.
  2. Ephemeral worker nodes consume jobs and launch isolated Docker containers or microVMs (AWS Firecracker / gVisor).
  3. Containers are spawned with strict cgroup resource constraints (256MB RAM, 0.5 vCPU), read-only root filesystems, and disabled network interfaces.
  4. Container stdout/stderr logs are streamed back via WebSocket gateway.
- **Code Reference**: services/runtimeService.js

I) CODE REVIEW QUESTIONS

Q21: Identify a code smell in `services/aiService.js` and how to fix it.

- **Difficulty Level**: Medium / Hard
- **Why Interviewer Asks**: Tests code quality evaluation and refactoring insight.
- **Ideal Answer**: In `aiService.js`, multiple functions (`aiCodeReview`, `aiCodeSuggestions`, `aiBugDetection`, `aiAutoFix`) duplicate repetitive JSON prompt wrapper boilerplate and Groq completion calls. Refactoring this into a generic helper function `callGroqTask<T>(taskType, schema, payload)` would reduce duplicate code by over 150 lines and centralize error parsing.
- **Code Reference**: services/aiService.js

J) LIBRARY AND DEPENDENCY QUESTIONS

Q22: Why is `archiver` used instead of `jszip` for project export?

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Tests library selection knowledge based on memory stream efficiency.
- **Ideal Answer**: `jszip` generates zip files in memory as complete Buffer objects before sending, which causes RAM spikes for large projects. `archiver` is built on Node's native Transform streams, allowing file data to be compressed and piped incrementally directly into the HTTP response stream (`archive.pipe(response)`) with minimal memory consumption.
- **Code Reference**: package.json & routes/platformRoutes.js

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K) DEPLOYMENT AND DEVOPS QUESTIONS

Q23: How are environment variables managed securely in production?

- **Difficulty Level**: Medium
- **Why Interviewer Asks**: Evaluates secret management practices.

- **Ideal Answer**: Sensitive keys (`JWT_SECRET` , `GROQ_API_KEY` , `FIREBASE_PRIVATE_KEY` , `RESEND_API_KEY`) are stored in AWS Secrets Manager or HashiCorp Vault and injected as environment variables at runtime via Docker container environment configs, never committed to git repositories (`.gitignore` includes `.env`).
- **Code Reference**: .env.example & .gitignore

L) PROJECT FAILURE SCENARIO QUESTIONS

Q24: What happens if MongoDB connection drops while the backend server is running?

- **Difficulty Level**: Hard
- **Why Interviewer Asks**: Tests error recovery and database connection retry logic.
- **Ideal Answer**: In server.js, `connectWithRetry()` catches MongoDB connection errors and retries every 5 seconds without crashing the server. The `/health` check endpoint returns `database: "disconnected"` with HTTP 200, allowing load balancers to detect degraded status.
- **Code Reference**: server.js & config/db.js

M) PERSONAL CONTRIBUTION QUESTIONS

Q25: What was the most challenging technical feature you implemented in DevWatchAI?

- **Difficulty Level**: Personal / Behavioral
- **Why Interviewer Asks**: Tests technical ownership and problem-solving resilience.
- **Ideal Answer**: "The most technical feature was real-time terminal output streaming from ephemeral process runtimes. I had to coordinate writing temporary files safely using `fs.mkdtemp` , spawning non-shell child processes with `child_process.spawn` , piping stdout/stderr chunks over Socket.io rooms, enforcing a 30-second execution kill timer, and cleaning up temporary disk directories on process exit without blocking the main Event Loop."
- **Code Reference**: services/runtimeService.js

N) TRICK QUESTIONS

Q26: Trick Question: "Why didn't you use WebSockets for AI Code Reviews instead of REST APIs?"

- **Difficulty Level**: Hard / Trick
- **Why Interviewer Asks**: Tests if you blindly use WebSockets everywhere or understand protocol fit.
- **Ideal Answer**: WebSockets are intended for bi-directional, high-frequency, multi-client state synchronization (like presence cursors or streaming logs). AI Code Reviews are request-response operations tied to a specific document payload. REST APIs with HTTP JSON responses fit standard request-response semantics better, allow standard middleware processing, and support caching and clear status codes (`200 OK` , `400 Bad Request` , `503 Service Unavailable`).
- **Code Reference**: routes/aiRoutes.js

Q27: Trick Question: "Is your Monaco Editor rendering server-side on Next.js?"

- **Difficulty Level**: Trick
- **Why Interviewer Asks**: Tests deep knowledge of SSR constraints.
- **Ideal Answer**: No, Monaco Editor cannot render on the server because it depends on client-side browser APIs (`window` , DOM canvas, Web Workers). We explicitly disable SSR using Next.js `dynamic(() => import('@monaco-editor/react'), { ssr: false })` .
- **Code Reference**: components/cloud-ide.tsx

15. RESUME-BASED INTERVIEW QUESTIONS & PORTFOLIO PROJECT DEEP-DIVES

Candidate Resume Portfolio Overview

Based on your resume, senior interviewers will evaluate your technical depth across your three flagship projects:

1. **DevWatchAI**: AI-Powered Developer Collaboration Platform (*Next.js, TypeScript, Express.js, Firebase, MongoDB Atlas*)
2. **AIRA**: AI Voice Assistant (*React, Vite, Tailwind CSS, Vercel*)
3. **PropVera**: Real Estate Management Platform (*React, Tailwind CSS, Firebase*)

Interviewers will cross-examine your technical choices, architectural decisions, real-time data synchronization patterns, and demand defense for every quantitative metric on your resume.

A) DEVWATCHAI RESUME BULLET DEFENSE & INTERROGATION

Bullet 1: "Engineered a full-stack AI-powered developer collaboration platform integrating 5+ core technologies across frontend, backend, authentication, and cloud data layers."

- **Interviewer Question**: "*What are the exact 5+ core technologies, and why was each strictly required rather than using a unified monolith like Next.js API routes?*"
- **Difficulty**: Medium / Hard
- **Interviewer Intent**: Tests if you actually understand the technology boundaries or just listed popular buzzwords.
- **Ideal Answer**:
  1. **Next.js 16 (Client UI)**: Selected for client-side component architecture, dynamic Monaco editor mounting, and responsive workspace layout rendering.
  2. **Express.js (Backend Node Server)**: Required because Next.js serverless functions cannot maintain long-lived stateful processes. Express manages Socket.io room instances and spawns child processes for code execution.
  3. **Firebase Auth + JWT**: Manages multi-provider OAuth (Google, Email/Password) while issuing custom signed 7-day JWT tokens for backend API authentication.
  4. **MongoDB Atlas**: Serves as the primary document database for dynamic directory file nodes, version snapshots, and activity audit logs.
  5. **Groq Cloud LLM SDK**: Provides ultra-fast inference (`llama-3.3-70b-versatile`) using structured JSON mode for automated code reviews and auto-fixes.
  6. **Socket.io**: Handles bi-directional, low-latency WebSocket presence, cursor tracking, and streaming terminal execution logs.
- **Code Reference**: server.js, components/cloud-ide.tsx, services/socketServer.js

Bullet 2: "Implemented 6+ major capabilities, including Google OAuth, JWT-based authentication, AI-powered code reviews, collaborative workspaces, cloud IDE functionality, and real-time team chat."

- **Interviewer Question**: "*Walk me through the exact authentication bridge between Firebase Google OAuth and your custom Express JWT session tokens.*"
- **Difficulty**: Hard
- **Interviewer Intent**: Evaluates security architecture knowledge and identity verification mechanics.
- **Ideal Answer**:
  1. Client authenticates via Firebase Web SDK (`signInWithPopup(auth, googleProvider)`).
  2. Client receives a Firebase ID Token and passes it in the header to `POST /api/auth/login` .
  3. Server invokes `admin.auth().verifyIdToken(idToken)` in config/firebaseAdmin.js.
  4. Upon verification, the server upserts a `User` document in MongoDB Atlas keyed by `firebaseUid` or `email` .
  5. Server signs a custom 7-day JWT containing `{ userId, uid, email }` using `jsonwebtoken` and `JWT_SECRET` .
  6. Client stores the JWT in `localStorage` and supplies it as a `Bearer` token for all subsequent REST and Socket.io operations.
- **Code Reference**: middleware/auth.js & routes/platformRoutes.js

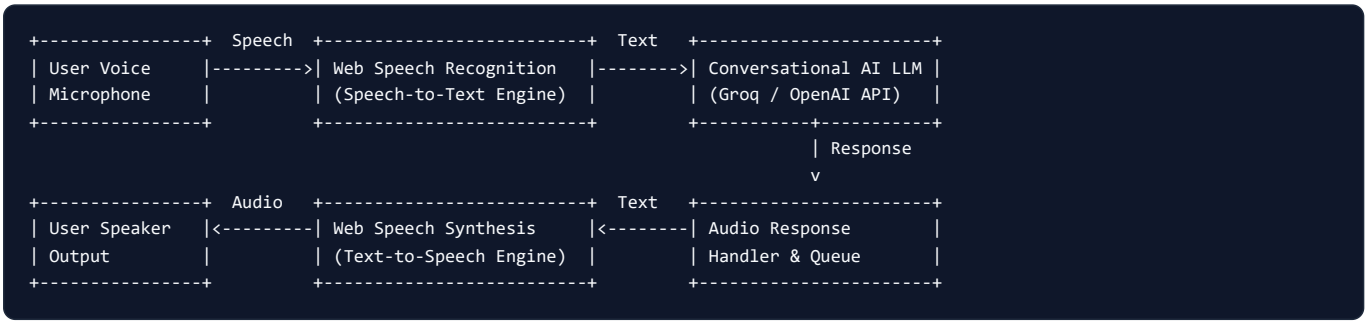
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B) AIRA - AI VOICE ASSISTANT DEEP-DIVE

Project Summary & Stack

- **Tech Stack**: React, Vite, Tailwind CSS, Vercel
- **Core Functionality**: Real-time voice interaction combining browser speech recognition (Web Speech API / SpeechToText), conversational LLM processing, and speech synthesis (TextToSpeech) for hands-free AI interaction.

Technical Architecture & Data Flow



Resume Questions & Ideal Answers for AIRA

Q1: "How did you implement real-time voice input without high latency or audio clipping?"

- **Difficulty**: Hard
- **Interviewer Intent**: Assesses browser multimedia API handling, stream management, and latency optimization.
- **Ideal Answer**:

> "I utilized the browser's native `webkitSpeechRecognition` / `SpeechRecognition` API configured with `continuous: false` and `interimResults: true`. As the user speaks, interim transcriptions update the UI in real time. Upon speech pause (`onresult` final event or silence detection timer), the full transcript string is dispatched immediately to an ultra-fast LLM endpoint (Groq / OpenAI). To prevent audio clipping during speech synthesis, I implemented an audio queue manager using `window.speechSynthesis` that cancels previous utterances (`speechSynthesis.cancel()`) when new priority voice responses arrive."

Q2: "How does AIRA compare structurally to DevWatchAI's AI assistant?"

- **Difficulty**: Medium
- **Interviewer Intent**: Tests cross-project architectural awareness and protocol comparison.
- **Ideal Answer**:

> "AIRA is a client-heavy, single-page voice processing web app hosted on Vercel. It relies primarily on browser native APIs (SpeechRecognition and SpeechSynthesis) and communicates directly with streaming LLM APIs. In contrast, DevWatchAI is a full-stack, stateful platform running a dedicated Node/Express backend, managing real-time WebSocket rooms, MongoDB version trees, and isolated subprocess execution. While AIRA optimizes for low-latency client-side voice streaming, DevWatchAI optimizes for multi-user collaboration and secure code sandboxing."

Q3: "What happens if a user's browser does not support the Web Speech API?"

- **Difficulty**: Medium
- **Interviewer Intent**: Tests graceful degradation and fallback UI engineering.
- **Ideal Answer**:

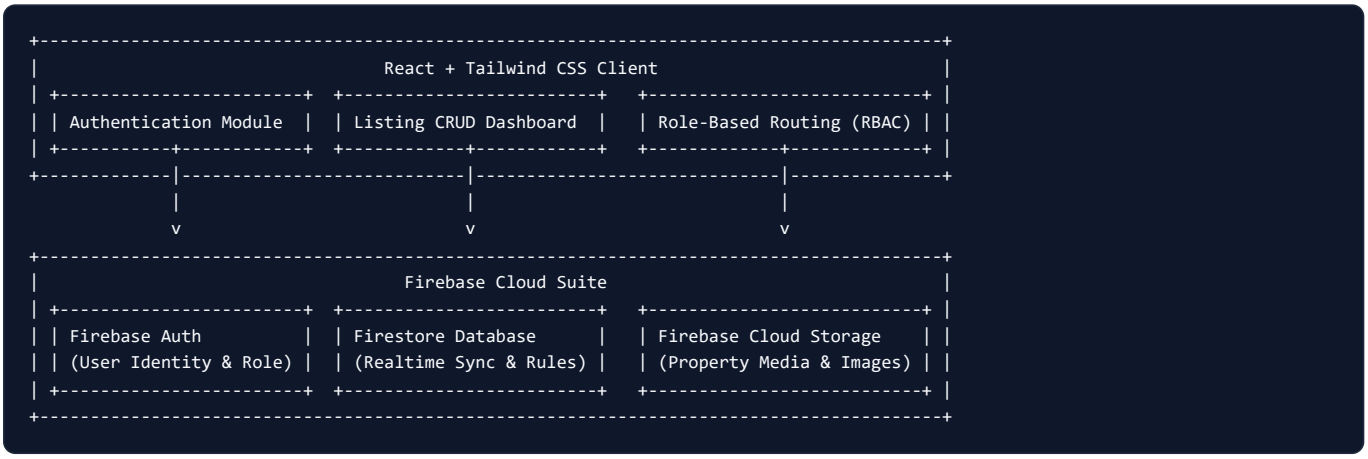
> "I built a feature-detection fallback. On component mount, the app checks `('SpeechRecognition' in window || 'webkitSpeechRecognition' in window)`. If unsupported (e.g. older Firefox versions), AIRA gracefully falls back to a text-input interface with a visual alert explaining that voice input is unavailable on the current browser, ensuring zero application crashes."

C) PROPVERA - REAL ESTATE MANAGEMENT PLATFORM DEEP-DIVE

Project Summary & Stack

- **Tech Stack**: React, Tailwind CSS, Firebase (Firestore, Auth, Storage)
- **Core Functionality**: Multi-user property listing and rental management platform featuring Role-Based Access Control (RBAC), property image uploads, real-time listing updates, and tenant/landlord dashboard management.

Technical Architecture & Data Flow



Resume Questions & Ideal Answers for PropVera

Q1: "How did you implement Role-Based Access Control (RBAC) in PropVera using Firebase?"

- **Difficulty**: Hard
- **Interviewer Intent**: Evaluates security architecture, token claims, and database authorization rules.
- **Ideal Answer**:

> "RBAC in PropVera is enforced at two levels:
> 1. **Client-Side Authorization**: User role metadata (`landlord` , `tenant` , `admin`) is stored in Firebase Auth custom user claims or user profile documents in Firestore. React Higher-Order Components / Protected Route wrappers restrict navigation routes (e.g. preventing tenants from accessing property creation forms).
> 2. **Database Security Rules**: In Firestore, security rules validate user identity and roles on every write operation:
>
> ```
> match /properties/{propertyId} {
>   allow read: if request.auth != null;
>   allow create, update: if request.auth.token.role == 'landlord' || request.auth.token.role == 'admin';
>   allow delete: if resource.data.ownerId == request.auth.uid;
> }
> ```
>
> This guarantees that client-side bypass attempts are blocked directly at the database layer."

Q2: "How does real-time database synchronization in PropVera differ from Socket.io synchronization in DevWatchAI?"

- **Difficulty**: Hard / Senior
- **Interviewer Intent**: Assesses knowledge of Firebase Firestore snapshot listeners vs custom WebSocket event broadcasting.
- **Ideal Answer**:

> "In PropVera, real-time sync is handled natively by Firebase Firestore using client-side snapshot listeners (`onSnapshot(docRef, callback)`). Firestore handles offline caching, delta synchronization, and WebSocket reconnection under the hood.
> In DevWatchAI, we built a custom WebSocket architecture using Socket.io on Express. We needed custom event channels (`presence:file` , `execution:log`) and fine-grained control over streaming process logs and cursor presence maps, which Firestore's document-oriented change stream model cannot handle efficiently."

Q3: "How did you handle image upload optimization for property listings in PropVera?"

- **Difficulty**: Medium
- **Interviewer Intent**: Tests client-side file validation, storage handling, and performance optimization.
- **Ideal Answer**:

> "Before uploading to Firebase Cloud Storage, images undergo client-side validation for file format (`jpg` , `png` , `webp`) and file size limits (max 5MB per image). Using browser Canvas APIs, high-resolution images are compressed on the client before upload, reducing storage bandwidth and accelerating page load times on the property detail views."

D) CROSS-PROJECT ARCHITECTURAL COMPARISON MATRIX

Architectural Dimension	DevWatchAI	AIRA	PropVera
Primary Domain	Collaborative Cloud IDE & AI Workspace	AI Voice Assistant	Real Estate Management Platform
Frontend Framework	Next.js 16 (App Router) + React 18	React 18 + Vite	React 18 + Vite / CRA
Backend Layer	Express.js (Dedicated Node Server)	Serverless / API Endpoints	Firebase Backend-as-a-Service (BaaS)
Database & Storage	MongoDB Atlas (Mongoose ORM)	N/A (Stateless Session State)	Firebase Firestore + Cloud Storage
Realtime Protocol	Socket.io (Custom WebSockets)	Web Speech API Streaming	Firestore WebSocket Snapshot Listeners
AI Integration	Groq SDK (`llama-3.3-70b-versatile`)	Web Speech STT/TTS + LLM API	N/A
Authentication	Firebase Auth -> Express Custom JWT	N/A / Web Browser Session	Firebase Auth (Custom Claims RBAC)
Hosting & Deployment	Node Server (Render / AWS / Docker)	Vercel (Static Web App)	Firebase Hosting / Vercel

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E) BEHAVIORAL & RESUME METRIC DEFENSE STRATEGY

1. How to Defend Resume Bullet Numbers

When an interviewer points to specific numbers on your resume:

- **"Integrating 5+ core technologies"**: List Next.js, Express.js, Firebase, MongoDB Atlas, Groq SDK, Socket.io. Explain the boundary of each.
- **"Implemented 6+ major capabilities"**: Google OAuth, JWT authentication, AI code reviews, collaborative workspaces, Monaco Cloud IDE, real-time team chat.
- **"4+ modern web technologies" in AIRA**: React, Vite, Web Speech API (SpeechRecognition + SpeechSynthesis), Tailwind CSS.
- **"3 core AI capabilities" in AIRA**: Speech Recognition (STT), Conversational Reasoning (LLM), Speech Synthesis (TTS).
- **"4 key modules" in PropVera**: User Authentication & RBAC, Property Listing Management (CRUD), Real-time Synchronization, Media Upload & Storage.

2. Master Pitch Connecting All 3 Projects

> *"Across my projects, I've specialized in building high-performance, real-time, interactive web applications. **PropVera** gave me deep experience in multi-user workflows, role-based security, and cloud database synchronization. **AIRA** allowed me to push the boundaries of browser multimedia APIs, streaming voice input, and low-latency AI interactions. Finally, **DevWatchAI** represents the synthesis of these skills: a full-stack collaborative platform combining stateful Express backend servers, custom Socket.io real-time presence engines, sandboxed code execution, and Groq-powered AI code auditing."*